

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2017
CE 2241
(Principles of Environmental Engineering)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable values for any missing data.

Section – A

1. (a) What do you mean by environmental engineering? Define ABIOTIC and BIOTIC environment. What is the interrelation between them? (09)
(b) What are the objectives of water supply? What is meant by potable water and palatable water? (09)
(c) Mention the criteria required for water quality. Write down the components of water supply system with neat sketches. (12)
(d) What do you mean by the term "safe and wholesome" water? (05)
2. (a) Write down the sources, standard values and significance of the following water quality parameters: (i) Colour, (ii) Hardness, (iii) Iron (iv) Arsenic (10)
(b) What is coagulation? Which are the widely used coagulants? Mention a comparative figure of two coagulants and also describe their benefits. (10)
(c) Find the dimension of a settling tank to treat 40 m³ of raw water per hour when the detention time is 2 hours. (10)
(d) Write down the unit operations available for removal of specific impurities. (05)
3. (a) Define filtration. Why are roughing filters constructed? Write down the types of roughing filters with neat sketches. (15)
(b) What are the methods available for arsenic removal from groundwater? Write a short note on lime treatment of water. (08)
(c) The population of a community in 1950, 1960, 1970, 1980, 1990 and 2000 was 300, 340, 380, 420, 450 and 500 respectively. Estimate the population of the community in the year 2010 and 2020 by least square parabola method. (12)
4. (a) What do you mean by fire demand and water hammer? Write down the qualities required for pressure pipes. (10)
(b) In constructing an elevated circular R.C.C tank, if the cost per sft of the shell is 2 times to that of the floor, what will be the most economical dimensions of the tank? (10)
(c) Describe the influencing factors of the deterioration. Write down the fate and origin of the following air pollutants - (i) Carbon Monoxide, (ii) Lead (iii) Nitrous Dioxide. (15)

Section – B

5. (a) What is meant by sanitation? Write down the principle objectives of sanitation. (05)
(b) Define Sanitation System with its different functions. (05)
(c) Distinguish between on-site and off-site sanitation system. Describe the suitability of sanitation systems based on water consumption and population density. (13)
(d) Design a leach pit for both single and alternating twin off-set pit pour flush latrines serving a family of ten members living in a rural area. Wastewater flow is 10 lpcd and soil is clay type. (12)

6. (a) Write down the problems associated with traditional simple pit latrines. How can these problems be minimized? Describe briefly. (08)
- (b) Define septic tank. Show the different components of a septic tank with neat sketch. Write down the processes take place within the septic tank. (12)
- (c) A septic tank is to be designed to serve a residential building having twelve residences. Wastewater generation rate is 80 lpcd. The tank is to be desludged every three years and minimum mean hydraulic detention time for sedimentation is 0.58 days. (15)
7. (a) Write down the functions of an inlet and outlet device. Why the inlet of a septic tank is usually made with a 'tee' Junction? (05)
- (b) What is meant by Waste Stabilization Pond? Why optimum retention time in an anaerobic Pond is 5 days? (05)
- (c) State and proof Marais' theorem about the maximum efficiency in a series of waste stabilization pond. (08)
- (d) Design a waste stabilization pond system to treat 12,000,000 litre/day of domestic sewage having BOD_5 700 mg/L; Design temperature = 20°C. Required effluent $BOD_5 < 30$ mg/L; FC < 5000/100 ml. Given: $N_l = 4 \times 10^7$ /1000 ml. (17)
8. (a) Describe the symbiosis between Algae and Bacteria. (06)
- (b) What is meant by conventional wastewater treatment? Show the flow diagram of Activated Sludge Process. (06)
- (c) Distinguish between sewer, sewage and sewerage. (06)
- (d) What is meant by Solid Waste Management? What are the objectives of Solid Waste Management? (07)
- (e) Write down the advantages of different conventional wastewater collection system. (10)
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Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 2nd Year 2nd Term Examination, 2017
HUM 2217
(Public and Local Government Finance)

Full Marks: 210

Time: 3.00 hrs

- N.B. i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) What is public finance? Describe the importance of studying Public Finance in the department of Urban and Regional Planning? (15)
(b) Charlie loves to see his baby watching *Teletubbies* on his local public TV station; but he never sends money to support the station during their fund-raising drives. What name economists have for Charlie? How can government solve the problem caused by the people like Charlie? Explain. (20)
2. (a) Greater consumption alcohol leads to more motor vehicle accidents and thus imposes cost on people who do not drink and drive. Illustrate market for alcohol, labeling the demand curve, the social-value curve, the supply curve and the social cost curve. (15)
(b) Explain the market equilibrium level of output and the efficient level of output. On your graph, shade the area corresponding to the deadweight loss of the market equilibrium. (Hint: The deadweight loss occurs because some units of alcohol are consumed for which the social cost exceeds the social value) Explain. (20)
3. (a) Define price ceiling and price floor, and give an example of each. Which leads to a shortage? Which leads to a surplus? Why? (15)
(b) Consider the market for rubber bands. If this market has very elastic supply and very inelastic demand, how would the burden of a tax on rubber bands be shared between consumer and producer? Use the tools be consumer surplus and producer surplus in your answer. (20)
4. (a) What are the factors on which incidence of tax depends? Trace the incidence of estate duty and inheritance tax. (15)
(b) Define tax. What are the characteristic of a good tax system? (10)
(c) "The principle of equity necessary implies least aggregate sacrifice for the society" Discuss. (10)

Section – B

5. (a) What is government budget? Explain importance of government budget. (20)
(b) Discuss the steps of formulating a budget. (08)
(c) What are the components of government budget? Explain. (07)
6. (a) What are the objectives of public expenditure? How far dose public expenditure in Bangladesh fulfils those objectives? (20)
(b) Discuss the principle of government expenditure. (08)
(c) What will be the government expenditure for full employment level income 1500 unit? Where investment expenditure 50 unit and consumption function $c = 100 + 0.25y$ (07)
7. (a) How a union parishad formed its fund and charged expenditure. (20)
(b) Explain property rules of union parishad. (08)
(c) How the local government like union parishad prepare and implement its development plan. (07)
8. (a) What is local government? Explain the structure if local government of Bangladesh. (15)
(b) What are the main sources of revenue of local Government of Bangladesh? (10)
(c) Explain the role of deficit Financing in Lower-Middle class Income Country (LMIC) like Bangladesh. (10)

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 2nd Year 2nd Term Examination, 2017
URP 2251



(Landscape Planning and Design)

Full Marks: 210

Time: 3.00 hrs

- IV.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) What do you understand by Landscape Planning? Briefly discuss the importance of studying Landscape Planning and Design by Urban and Regional Planners. (10)
(b) Define the following terms with examples: (15)
(i) Landscape features (ii) Landscape elements (iii) Landscape Principles.
(c) Draw a neat sketch of designed Landscape Plan showing the use of different types of lines. (10)
2. (a) Briefly discuss the factors to be considered in site planning with appropriate examples of each. (20)
(b) Mention the important properties of texture as a landscaping element. Provide a Texture Study to elucidate the fact. (15)
3. (a) Write down with examples the significance of ecology for man from different perspectives of landscaping planning and design. (15)
(b) Discuss the Nitrogen Cycle with necessary diagram. (10)
(c) Mention the points to be considered while considering a landscape design for an industrial area. (10)
4. Annotate the following (any five): (35)
(a) Landscaping with shrub and ground cover forms (b) Landscaping with native wildflowers and grasses (c) Asymmetrical Balance (d) Repetition in Landscaping (e) Unity of Tree (f) The Biosphere (g) The Carbon Cycle

Section – B

5. (a) What is Landscape Planning? (05)
(b) Explain Contemporary Landscape design trend with examples. (15)
(c) What is landscape site? Explain the site inventory and analysis process. (15)
6. Write short note on following topics with necessary sketches and examples- (35)
(a) Soil Layer (b) Hardscape (c) Streetscape (d) Vista (e) Context (f) Urban Scale (g) Cascade
7. (a) What is Cityscape? Explain the city vibration pattern with necessary sketches. (15)
(b) Explain the following topics with sketches- (15)
(i) View, (ii) Axis, (iii) Terminus
(c) What is landscape conservation? (05)
8. (a) Explain the Residential Landscape Planning and Design steps with necessary sketches. (15)
(b) Discuss about water edge landscape principles with necessary sketches and examples. (10)
(c) What is Buffer zone? Explain with necessary sketches. (10)



URP-2281
(GIS and Remote Sensing)

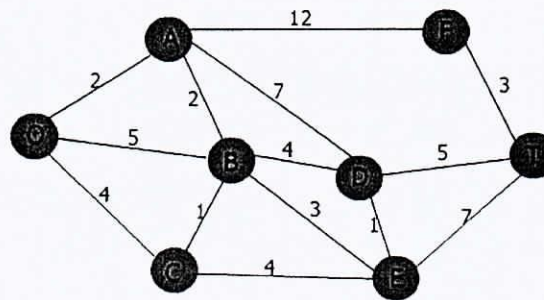
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Show the relationship between Spatial Data Modeling and Database Management System. (05)
- (b) Write short notes on the following: (15)
 - (i) IDW (ii) Semivariogram (iii) The exponent of distance
- (c) Solve the shortest path from O to T by applying the Dijkstra's Algorithm on the following network. Also solve each node. (15)



2. (a) Mention the Classic Network-type problems (05)
- (b) Briefly explain the following: (15)
 - (i) The travelling salesperson problem (ii) Thiessen Polygon and its construction (iii) Path finding
- (c) Provide examples of GIS applications in major sectors with respect to Bangladesh. (10)
- (d) What do you understand by layers in GIS? (05)
3. (a) Provide information about the following types of map: (15)
 - (i) Agro-ecological Zone (ii) Inundation Land Type (iii) Crop Suitability
- (b) What is a spatial interpolation? Show the difference between Deterministic methods and Geostatistical methods. (10)
- (c) Show the errors in digitizing with the help of appropriate diagrams. (10)
4. (a) Mention the components of GIS. Why are people both providers and users? (10)
- (b) Explain the following types of spatial queries/operations with the help of a continuous example. (10)
 - (i) Containment Query (ii) Clipping (iii) Spatial Join (iv) Map Overlay (v) Merge/Aggregation
- (c) Briefly describe the ArcGIS Desktop Modules. (10)
- (d) What do you understand by "Geodatabase"? (05)

Section – B

5. (a) Describe the relationship among three classifications (High, Medium and Low) of each type of resolutions (spatial, spectral, radiometric and temporal). (10)
- (b) Explain the relations between Instantaneous Field of View (IFOV) and Angular Field of View (AFOV) with providing proper figure. How does swath width of the satellite images depend upon the Angular Field of View (AFOV)? (10)
- (c) Write down the name of the corrections for satellite image processing (it may be pre-processing or post-processing) with proper explanation based on the following different types of error. (10)
- Terrain effect due to undulation on the earth surface
 - Absence of sufficient local solar illumination
 - Satellites or aircraft not staying at a constant altitude or by sensors deviating from the primary focus plane
 - Variation of the surface reflectance on the same surface due to bad weather
- (d) "High spatial resolution satellite Images do not reflect always good quality, rather it may depend on other types of resolutions especially radiometric resolutions" – explain with necessary example. (05)
6. (a) "A common radiometric response is required for quantitative analysis of multiple satellite images of a scene acquired on different dates with different sensor." – are you agreed with the statement? State your rationales with your answer. (10)
- (b) Write down the different types of satellite image processing. Provide your rationales about the spatial and radiometric enhancement of remote sensing Images. (10)
- (c) Compare between two sets of satellite Images having different types of different resolutions in the following. Explain with proper example. (10)
- | Satellite Image (A) | Satellite Image (B) |
|---|--|
| <ul style="list-style-type: none"> High spatial resolution (0.3 - 4 m) Low spectral resolution (1- 3 bands) High radiometric resolution (16 bit and above) | <ul style="list-style-type: none"> High spatial resolution (0.3 - 4 m) High spectral resolution (16- 220 bands) Medium radiometric resolution (8 bit image) |
- (d) Explain Electromagnetic Spectrum in case of Remote Sensing with necessary figures. (05)
7. (a) Differentiate between Photogrammetry and Remote Sensing technology. Why are aerial photographs called the first form of remote sensing image? (10)
- (b) Explain the relationship between photo scale and map scale with proper figure. (05)
- (c) Briefly explain the different types of scale of a vertical photograph by drawing proper figure. (10)
- (d) Assume that the distance between two points was measured to be 120 mm on a vertical photograph and 300 mm on a map. If the surveying ground distance between the same two points is 6000 m, what will be the photo scale and map scale and the scale reciprocals of the photograph and the map? Provide your judgments on the photo scale and map scale to describe the level of detailing of the photographs. (10)
8. (a) Explain the different types of remote sensing platforms along with their respective distinct applications. (10)
- (b) Explain any three applications of Photogrammetry of the following: (15)
- Exploration and Reconnaissance
 - Hydrographic Maps and Charts
 - Land Use Maps
 - Cadastral Plans
- (c) "Photography is the first part of photogrammetry where metrology is the second part" – Briefly explain with proper example. (05)
- (d) Write down the disadvantages of Photogrammetry. (05)